

Diffusion and Dissolution

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Out-line

- Introduction
- Solute transport
- Fick's laws of diffusion
- Non steady state and steady state diffusion and Lag time
- Diffusion through a membrane
- Distribution or partition coefficient
- Diffusion coefficient and permeability coefficient
- Dissolution
- Diffusion Controlled drug Delivery system
- Dissolution and Diffusion Controlled drug Delivery system
- Osmotic controlled drug delivery system

Objectives

- ❑ After the end of this topic ,students will be able to :
 - ❑ Know the different types of solute transports
 - ❑ Describe non-steady state and steady state diffusion
 - ❑ Calculate distribution or partition coefficient
 - ❑ Discuss dissolution processes
 - ❑ Understand how the diffusion, dissolution and osmotic controlled drug delivery systems work.

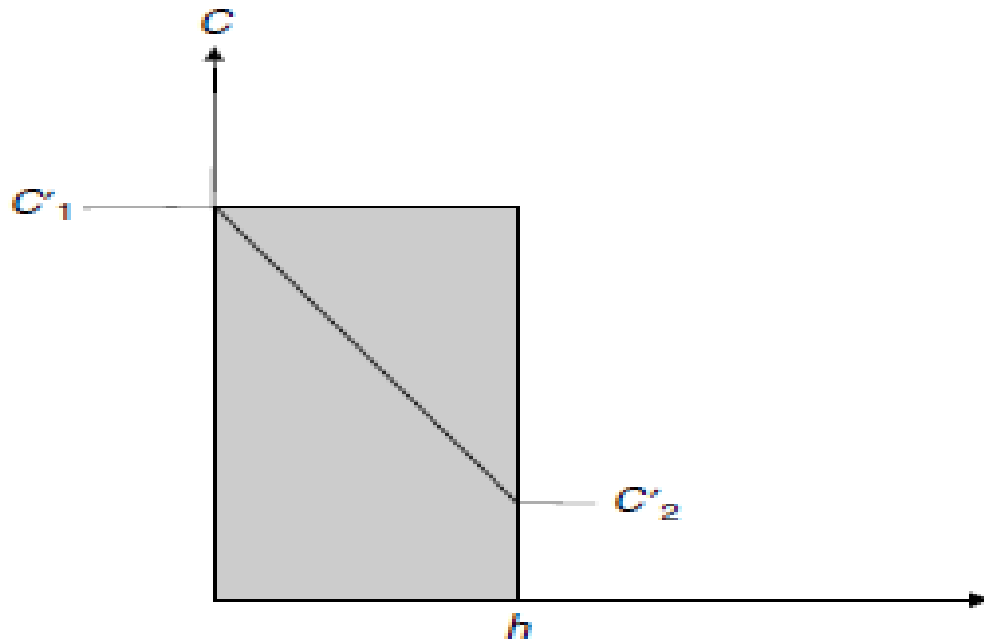
Introduction

- The mass transfer of a solvent (eg water) or a solute (eg a drug) forms a basis for many important phenomena in the pharmaceutical sciences
- **Diffusion** can be defined as a process by which molecules transfer spontaneously from a region of **higher concentration** to a region of **lower concentration**.

Introduction...

- The molecules that migrate from one location to another are termed as *diffusants or permeants or penetrants*
- The medium in which the diffusant migrates is called the *diffusional barrier*.
- *The concentration gradient is the concentration profile of the diffusant in the diffusional barrier.*
 - ✓ The concentration gradient is *the driving force* for diffusion.

Introduction...



- The number of molecules that diffuse through a unit area of the diffusional barrier in a given time is termed as *flux, J* .
- *Flux is a measure of the rate of molecular diffusion*

Fig. 1 Schematic representation of a diffusion system

Introduction...

- Diffusion governs the transport of the great majority of drugs across various biological barriers after administration.
- The theory of diffusion has been used in
 - investigating the mechanism of drug transport
 - designing and development of various controlled or sustained release

Introduction...

- **Osmosis** defined as movement of solvent across a selectively permeable membrane
- Down its concentration gradient

Solute transport

- various nutrients like Carbohydrates, Amino acids, Vitamins and drugs are transported and absorbed into the blood by various mechanisms across semi permeable membrane

- Passive diffusion

- Carrier mediated transport

- ✓ Facilitated transport

- ✓ Active transport

- Cotransport

- Vesicular transport

Solute transport...

Passive diffusion

- Major process for absorption of majority of drugs
- Diffusion follows Fick's law
- The drug molecules diffuse from a region of higher concentration to a region of lower concentration till equilibrium is attained
- Rate of diffusion is directly proportional to the concentration gradient across the membrane

Solute transport...

Factors affecting Passive diffusion:

- Diffusion coefficient of the drug
- Thickness and surface area of the membrane
- Size of the molecule
- concentration of the drug (concentration gradients)

Solute transport...

Certain characteristic of passive diffusion can be generalized

- Down hill transport
- Greater the surface area & lesser the thickness of the membrane, faster the diffusion
- Equilibrium is attained when the concentration on either side of the membrane become equal

Solute transport...

- Greater the **membrane/ water partition coefficient** of drug, faster the diffusion
- Passive diffusion process is energy independent but depends on the **square root** of the molecular size of the drugs
- The mol. Wt. of the most drugs lie between 100 to 400 Daltons which can be effectively absorbed passively

Solute transport...

Carrier mediated transport

- Involves a carrier which **binds reversibly** with the solute molecules (drug) to be transported to yield the carrier solute (drug) complex
- The carrier drug complex **transverses across** the membrane to the other side where it **dissociates** to yield the solute molecule (drug) and carrier
- The carrier then returns to its original site to accept a fresh molecule of solute (drug)
- carrier-mediated absorption is **saturable**

Solute transport...

- The maximum transport rate will be determined by the concentration of carrier molecules and the speed with which they can shuttle material across the membrane, and not on the concentration of the molecules being transported

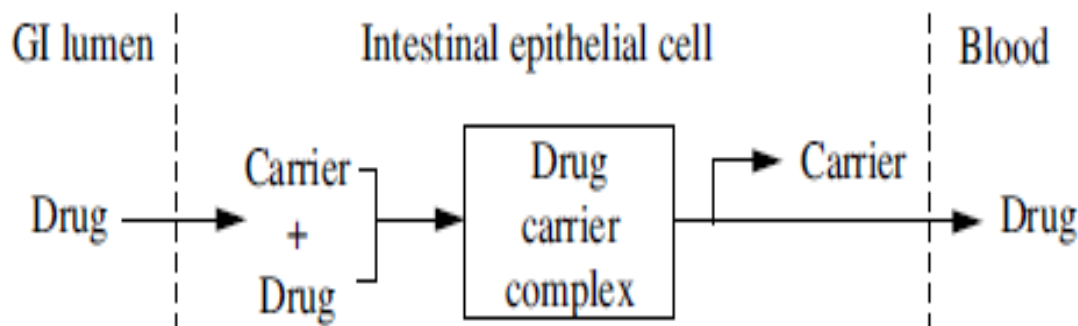
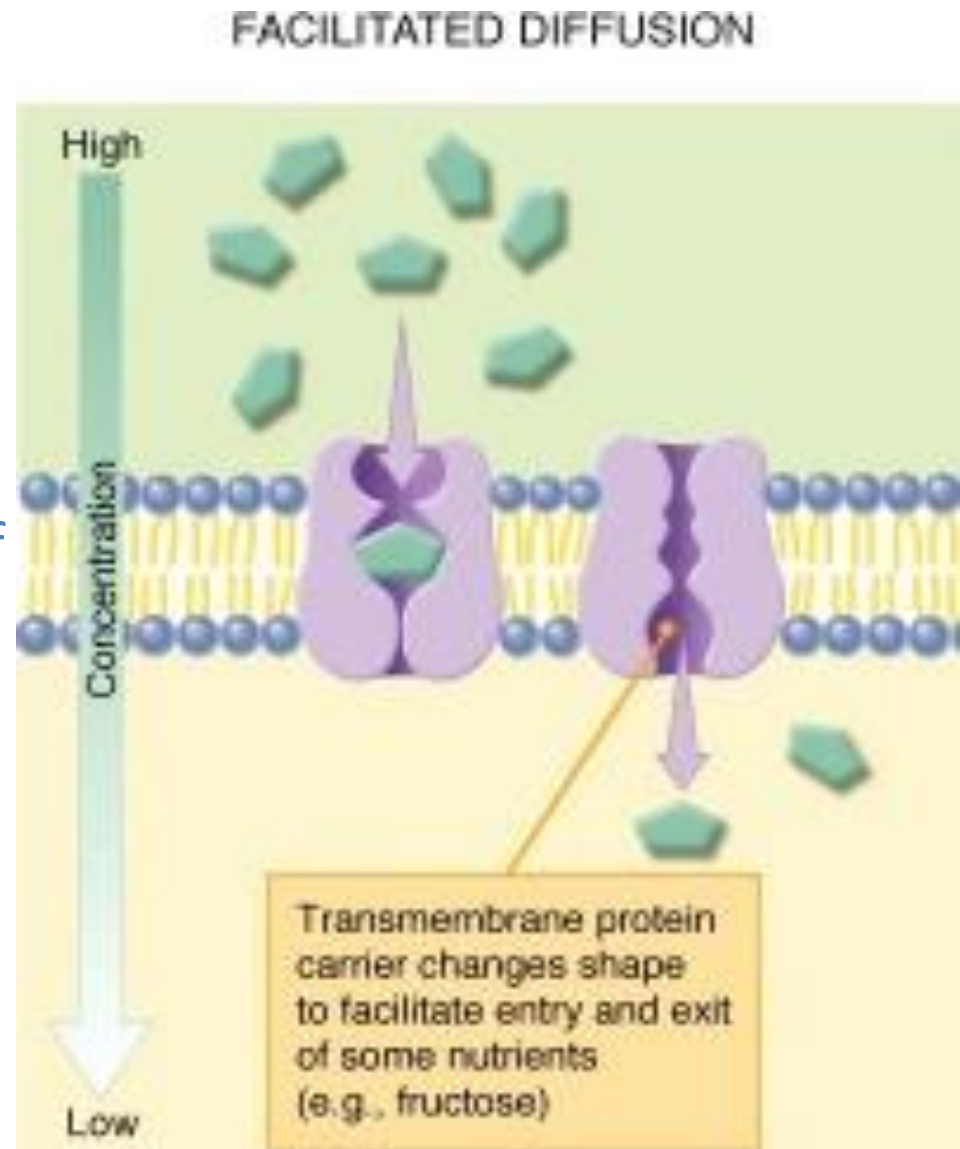


fig. Hypothetical carrier-mediated transport process

Solute transport...

Facilitated transport

- ✓ This mechanism involves the driving force as **concentration gradient**
- ✓ In this system, **no expenditure of energy** is involved (down-hill transport)
- ✓ the process is not inhibited by metabolic poisons that interfere with **energy production**.



Solute transport...

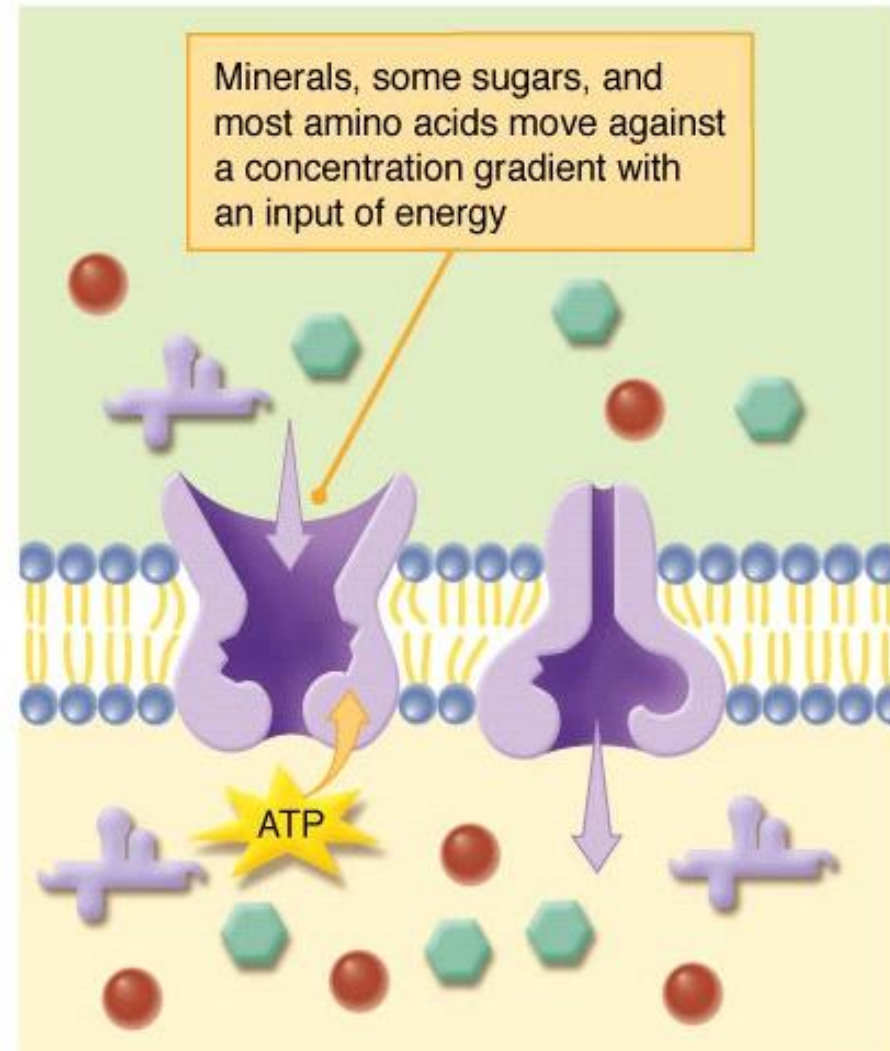
- Limited importance in the absorption of drugs.
e.g. Such a transport system include entry of glucose into RBCs & intestinal absorption of vitamins B₁, B₂ & B₁₂.
- An **intrinsic factor (IF)**, a glycoprotein produced by the gastric parietal cells, forms a complex with **vitamin B₁₂** which is then transported across the intestinal membrane

Solute transport...

Active transport

- Transport is against the concentration gradient (uphill transport)
- Since the process is uphill, energy is required
- As the process requires expenditure of energy
 - ✓ can be inhibited by metabolic poisons that interfere with energy production.

ACTIVE TRANSPORT



Solute transport...

- Substances that are transported actively include sodium, potassium, calcium, iron, glucose, amino acids , pyrimidine and vitamins like niacin, pyridoxine.
- Drugs having **structural similarity** to such agents are absorbed actively

Eg. Pyrimidine transport system – absorption of **5 FU**

L-amino acid transport system – absorption of
methyldopa and levodopa

Solute transport...

Cotransport

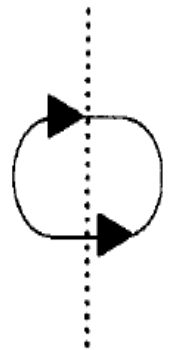
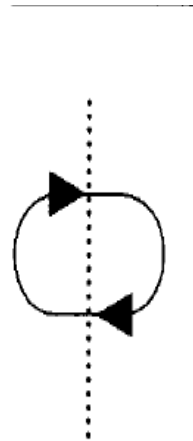
- Transport in which the cells' ion concentration gradient is used as a secondary energy source
- The transport proteins are systems which couple the transport of an ion to that of another molecule, so that, in allowing an ion to move out of the membrane to a lower concentration, the protein also moves a different molecule from lower to higher concentration.

Solute transport...

antiport

symport

- If the ion and molecule move across the membrane in the same direction, the process-
symport
- if they are exchanged in opposite directions -
antiport.



Solute transport...

- An important example of this process is the absorption of glucose from the intestinal lumen by the intestinal epithelial cells
 - This is accomplished by a sodium-glucose symport protein which couples the inward movement of a glucose molecule to that of a sodium ion

Drug transport system...

Vesicular transport (cytosis)

- Transport mechanism for the movement of particles into and out of cells.
- Pinocytosis - the engulfment of small solutes or fluid
- Phagocytosis - the engulfment of larger particles or macromolecules

Fick's law of diffusion

- Fick's law of diffusion govern diffusion processes in pharmaceutical systems

Fick's first law

- Fick's first law states that the rate of diffusion of solute molecule through a barrier or flux is proportional to the concentration gradient
- The flux , J , is the amount, M , of material flowing through a unit cross section, S , of a barrier in unit time, t .

$$J = dM / S \cdot dt \quad (1)$$

Fick's law of diffusion

- Since flux is proportional to the concentration gradient, dc/dx

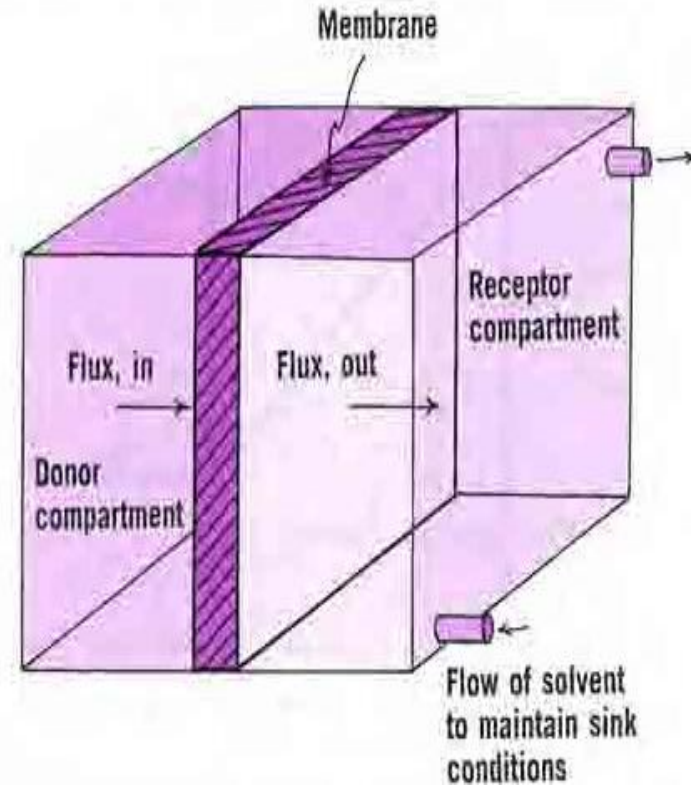
$$J = -D \frac{dc}{dx} \quad (2)$$

- ✓ D-diffusion coefficient of a penetrant (the diffusant)
(cm^2/sec)
- ✓ C-concentration in g/cm^3
- ✓ X- distance in cm of movement perpendicular to the surface
of the barrier
- ✓ Mass (gram or moles) , S (cm^2), time (sec)
- ✓ The unit of J is $\text{g}/\text{cm}^2 \text{ sec}$ and is positive quantity

Fick's law of diffusion...

Fick's second law

- states “the change in the concentration with time in a particular region is proportional to the change in the concentration gradient at that point in the system”.
- The diffusion equation is derived as follows
- The concentration, C , in a particular volume element changes as a result of net flow of diffusing molecules into and out of the region.



Fick's law of diffusion...

- The concentration of diffusant in the volume element changes with time, $\Delta C / \Delta t$, as flux changes with distance, $\Delta J / \Delta x$, in the x direction, or

$$\frac{\partial C}{\partial t} = - \frac{\partial J}{\partial x} \quad (3)$$

- Differentiating first law expression (eq. 2) with respect to x, one obtains

$$- \frac{\partial J}{\partial x} = D \frac{\partial^2 c}{\partial x^2} \quad (4)$$

Fick's law of diffusion...

- Substituting dc/dt from equation (3) in to equation (4) result in Fick's second law , namely

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 c}{\partial x^2} \quad (5)$$

- Eq. 5 represents diffusion only in the x direction
- To express concentration changes of diffusant in three dimensions, Fick's second law is written in the general form

$$\frac{\partial C}{\partial t} = D \left(\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial z^2} \right) \quad (6)$$

- Eq. 6 is not usually needed in pharmaceutical problems of diffusion

Non steady state and steady state diffusion

- **Nonsteady state diffusion** refers to a diffusion process in which the concentration of diffusant in a given space is a **function of time** or **the concentration** of diffusant in the diffusional barrier varies with time.
- In this case **Fick's second law is used** to study the diffusion process.
 - Mathematically, this condition can be described as

$$\frac{dC}{dt} = 0 \quad (7)$$

Non steady state and steady state diffusion...

- When the amounts of diffusant enter and leave the given space at the same rate, the concentration of the diffusant in the given volume is a constant and independent of time.
 - The diffusion process that meets this condition is considered a steady state diffusion.

$$\frac{dC}{dt} = 0 \quad (8)$$

- the concentration gradient is a constant for diffusion at the steady state
 - gives a constant flux

Time Lag (t_L)

- The time required for a diffusion to establish a uniform concentration gradient within the membrane that separates the donor and receptor compartments. Or
- The time required to attain the steady state.

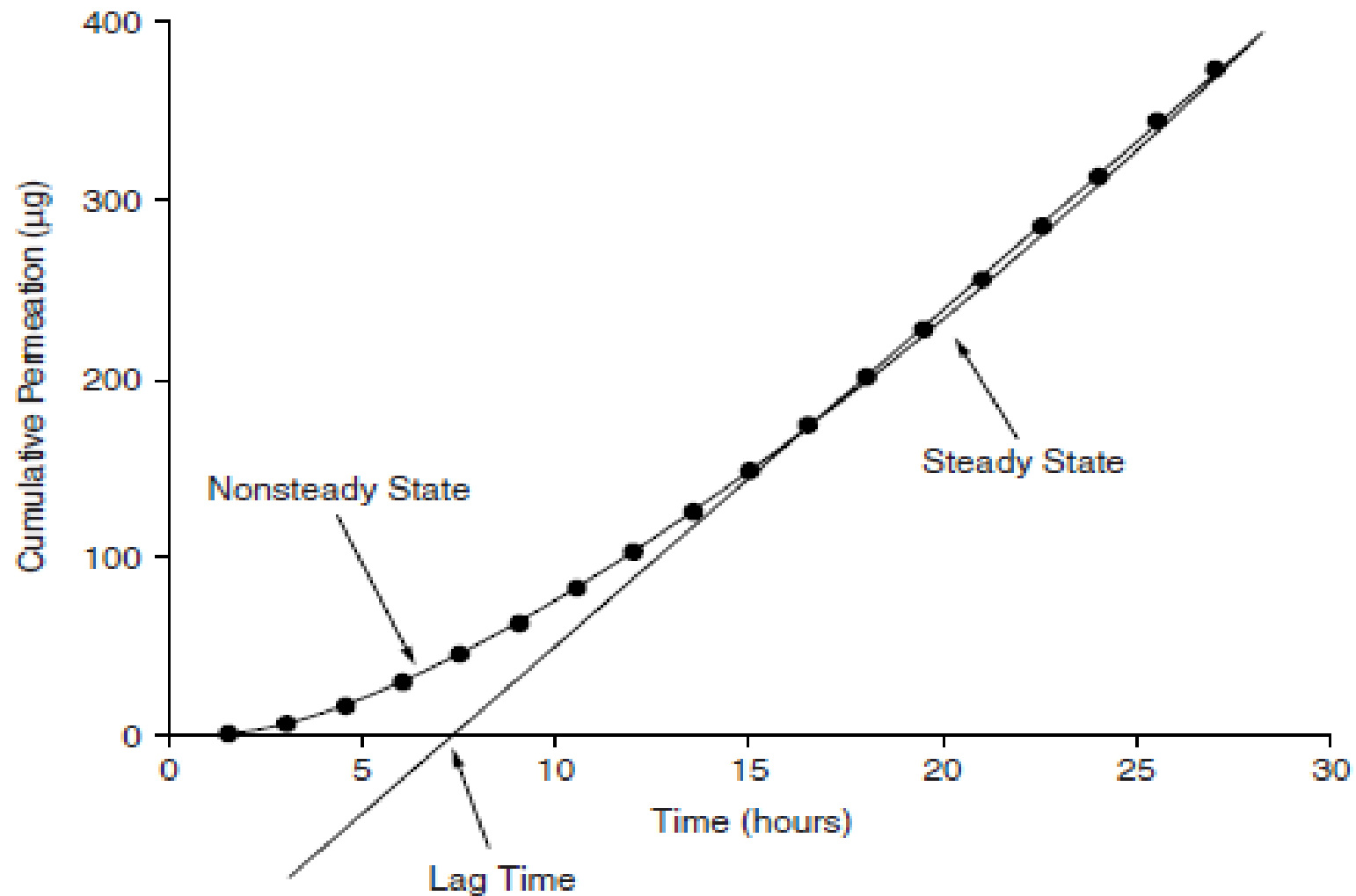
The lag time is given by:

$$t_L = h^2 / 6D \quad (9)$$

Where h is membrane thickness and D is diffusivity

- t_L may be obtained by extrapolating the steady state portion of the line to the time axis

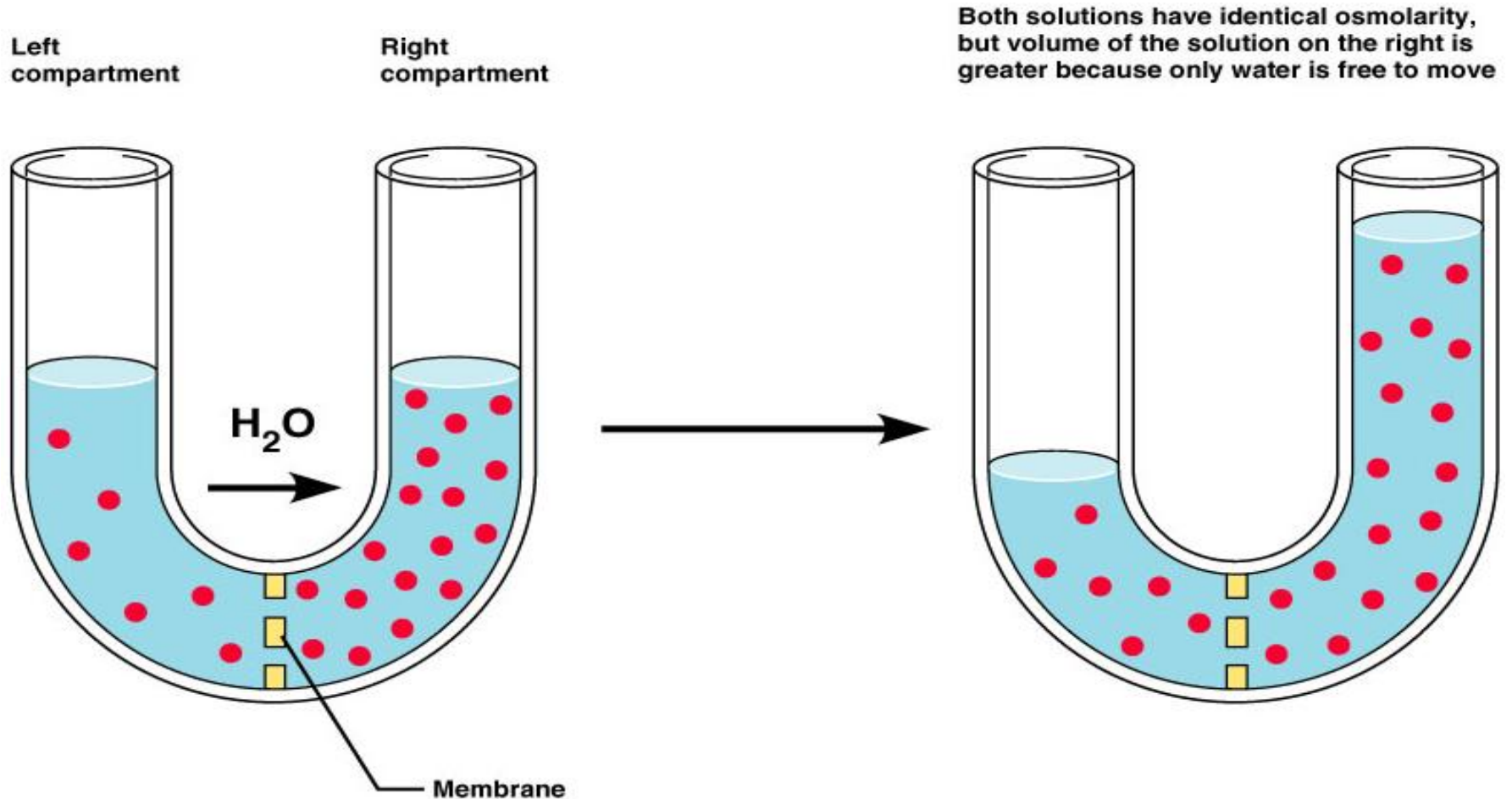
Time Lag (t_L)...



Osmosis

- Movement of solvent across a selectively permeable membrane
- Down its concentration gradient
- Toward the solution containing the higher solute concentration
 - This solution has a lower water concentration
 - Continues until solvent concentrations and solute concentrations are the same on either side of the membrane

Osmosis...



(b) Membrane impermeable to solute molecules, permeable to water

Diffusion through a membrane

- **Membrane** is a biological or physical “film” separating the phases and material passes by passive, active or facilitated transport across this film
- **Barrier** applies in a more general sense to the regions that are **resistance to passage** of a diffusing material.
- The **total barrier** being the sum of individual resistances of membranes.

Diffusion through a membrane...

Types of membrane

- **Impermeable membrane** - membrane through which nothing can pass
- **Freely permeable membrane** - many substance can pass through it
- **Selectively permeable membrane** - permits free passage of some materials and restricts passage of others

Diffusion through a membrane...

- Plasma membrane is selectively permeable
 - Distinction may be based on size, electrical charge, lipid solubility
- Cells differ in their permeabilities; depending on
 - what lipids and proteins are present in the membrane and
 - how these components are arranged

Diffusion through a membrane...

- Diffusion through lipid bilayer
 - Non polar, hydrophobic substances diffuse through lipid layer; these are “lipid soluble” or *lipophilic* substances

Diffusion through a membrane...

- Diffusion through channel proteins
 - hydrophilic and charged solutes diffuse through channel proteins; these are lipid insoluble or *lipophobic* (fat fearing) substances

Distribution or partition coefficient

- Distribution or partition coefficient is a measure of the ability of a compound to distribute in two immiscible phases.
- The partition phenomenon is of paramount importance for the diffusion across **skin and other epithelia**.

Distribution or partition coefficient...

- Many pharmaceutical processes based on the partition principles
 - ✓ absorption from the gastrointestinal tract after oral administration
 - ✓ drug distribution following entry into systemic circulation,
 - ✓ extraction and isolation of pure drugs after synthetic manufacturing or from crude plant sources,
 - ✓ formulation of a stable dosage form (emulsion, etc.)

Distribution or partition coefficient...

- The ability of drugs to penetrate a biological membrane can be evaluated using its partition in an **octanol and water system**.
- Occasionally, other organic solvents such as **chloroform, ether, and hexane** have been used as a lipid vehicle.
- When a drug is placed in an immiscible system composed of **octanol and water**,
 - ✓ the drug distributes in each solvent and eventually reaches equilibrium.

Distribution or partition coefficient...

- The ratio of drug concentration in each phase is termed its *distribution coefficient or partition coefficient (K)*

□ expressed as:

$$K = \frac{\text{Concentration in Octanol, } C_o}{\text{Concentration in Water, } C_w} \quad (10)$$

Practically, concentration in an **aqueous phase** is determined by chemical assays, such as **high performance liquid chromatography**, **ultra violet spectroscopy**, **gas chromatography**, **gas chromatography–mass spectroscopy**, etc.

Conce. In **octanol** is obtained by subtracting conc in aq from total conc.

Distribution or partition coefficient...

Q1: Succinic acid (0.15 g) dissolved in 100 ml of ether was shaken with 10 ml of water at 37°C. After equilibrium was achieved, the water layer contained 0.067 g of succinic acid. What is the partition coefficient (K) of succinic acid?

Q2: When boric acid is distributed b/n water and amyl alcohol at 25 °C, the concentration in water is found to be 0.0510 mole/liter and in amyl alcohol it is found to be 0.0155 mole/liter. What is the distribution coefficient?

Diffusion coefficient and permeability coefficient

- A diffusion coefficient represents the mobility of a molecule in a specific medium, the diffusional barrier.
- The mobility of a substance in a diffusional barrier is determined by:
 - ❖ the physicochemical properties of the diffusant
 - ❖ the diffusional membrane
 - ❖ temperature.

Diffusion coefficient and permeability coefficient...

- The relationship of the diffusion coefficient and these factors is expressed quantitatively in the Stokes–Einstein equation,
 - ✓ which states the diffusion coefficient as a function of temperature (T), viscosity (η), and size of the diffusant (r)

$$D = \frac{kT}{6\pi\eta r}$$

Diffusion coefficient and permeability coefficient...

- A set-up for studying diffusion through a diffusional membrane

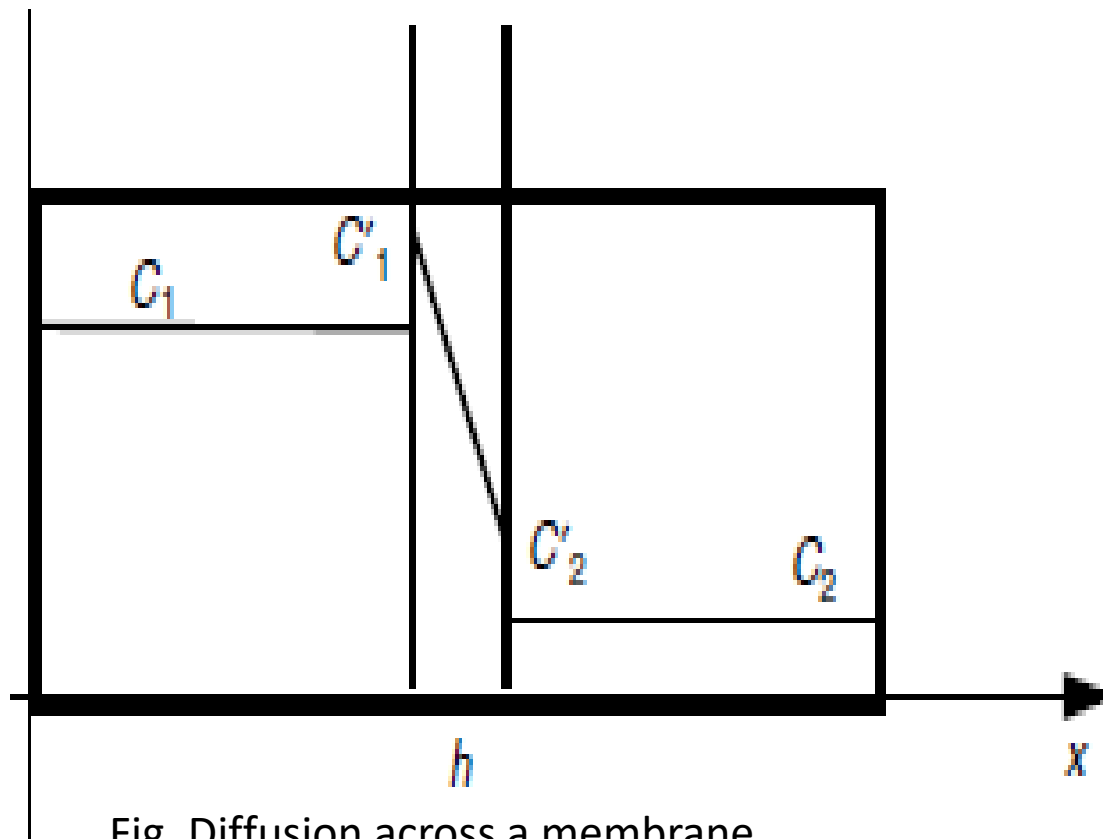


Fig. Diffusion across a membrane.

Diffusion coefficient and permeability coefficient...

- The set-up has two compartments divided by the diffusional barrier, for example, a membrane with a thickness of h .
- *The concentrations of diffusant outside of a diffusional barrier or the concentrations of diffusant in each compartment are denoted C_1 and C_2 .*
- *The concentrations of diffusant in the diffusional barrier are denoted C'_1 and C'_2 for each side of the diffusional barrier .*

Diffusion coefficient and permeability coefficient...

- *C₁ and C₂* can be determined experimentally, but *C'₁ and C'₂* are usually not known.
- To apply *Fick's laws*, one must know *C'₁ and C'₂*.
- The concentrations of a *diffusant* in the diffusional membrane and the adjacent medium can be related by using partition coefficients (*K*) as follows:

$$K = \frac{\text{Conc. in membrane}}{\text{Conc. in solution phase}} = \frac{C'_1}{C_1} = \frac{C'_2}{C_2}$$

Diffusion coefficient and permeability coefficient...

- Therefore, the concentrations of **diffusant** in the diffusional membrane can be expressed as:

$$C'_1 = KC_1 \text{ and}$$

$$C'_2 = KC_2$$

By replacing the concentrations of **diffusant** in the membrane by using these relationships, **Fick's first law** can be rewritten as

$$J = -\frac{DK}{h}(C_2 - C_1)$$

*D, K, and h are constant for a given diffusional barrier
a **permeability or permeability coefficient** (P), is defined as:*

$$P = \frac{DK}{h}$$

Diffusion coefficient and permeability coefficient...

- The dimension of permeability coefficient is length per unit time,
 - ✓ for example, centimeters per second.
- Permeability coefficient is commonly used in pharmaceutical sciences to determine or compare the permeation of a drug through various biological barriers.

Problem 1. A drug passing through a 1-mm-thick membrane has a diffusion coefficient of $4.23 \times 10^{-7} \text{ cm}^2 \text{ per sec}$ and an oil–water partition coefficient of 2.03. The concentration of the drug in the donor compartment is 0.5 mg/mL.

- a) Calculate the permeability and
- b) the diffusion rate of the drug or flux considering sink condition.

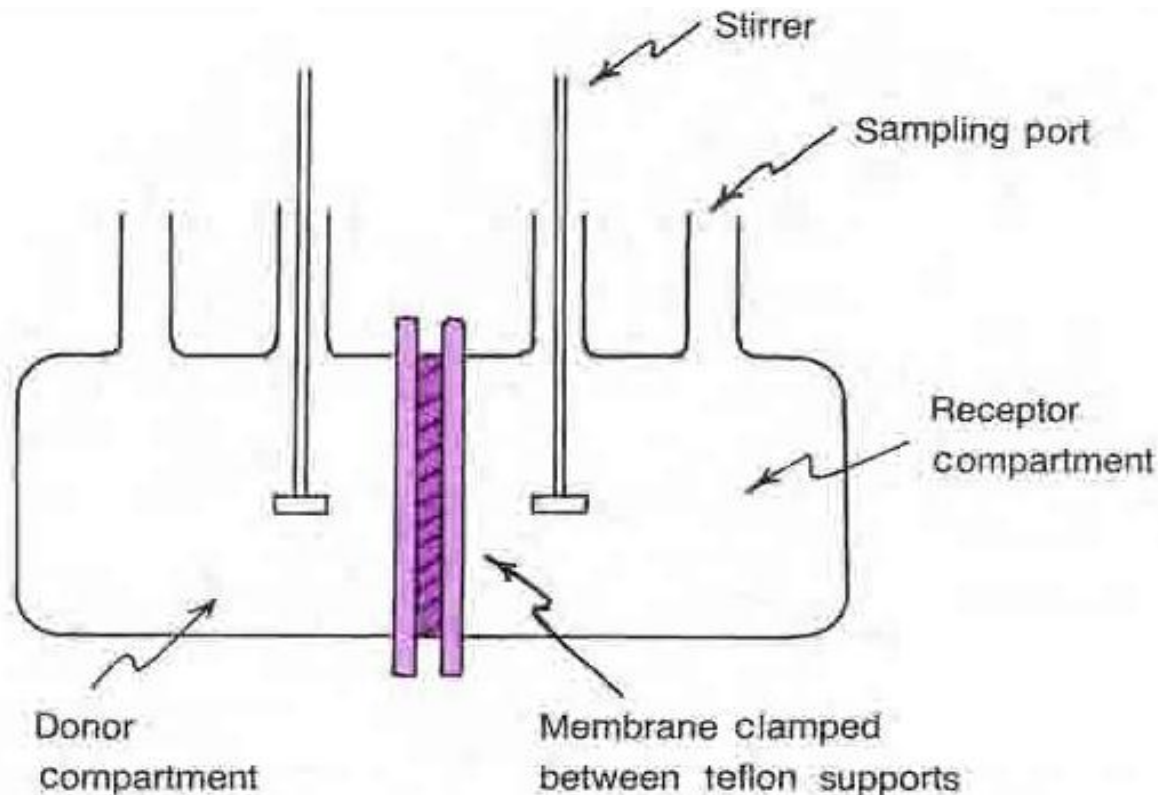
Problem 2. A steroid permeates through a 0.075 cm-thick membrane with a diffusion coefficient of $4.23 \times 10^{-7} \text{ cm}^2 \text{ per sec}$ and an oil–water partition coefficient of 2.03 . The concentration of steroid in the donor chamber is 0.004 mmol/ml and receptor chamber is 0.0001mmol/ml.

- (a) Calculate the permeability coefficient
- (b) Calculate the diffusion rate of the drug or flux

Diffusion study

- A number of apparatus have been described for studying the diffusion of drug across membranes

A simple diffusion cell



Diffusion study...

- The apparatus consists of two cells A and B separated by a barrier through which diffusion can occur
- The drug solution is in the DC while the solvent is in the RC
- Sample solutions are collected periodically from the RC and analysed by appropriate analytical methods
- Glass or clear plastic are easy to assemble and clear
 - This allows for the visualization of the liquid and of the rotating stirrer
- The chambers can be thermostated and automatic sampling devices can also be attached to them

Diffusion study...

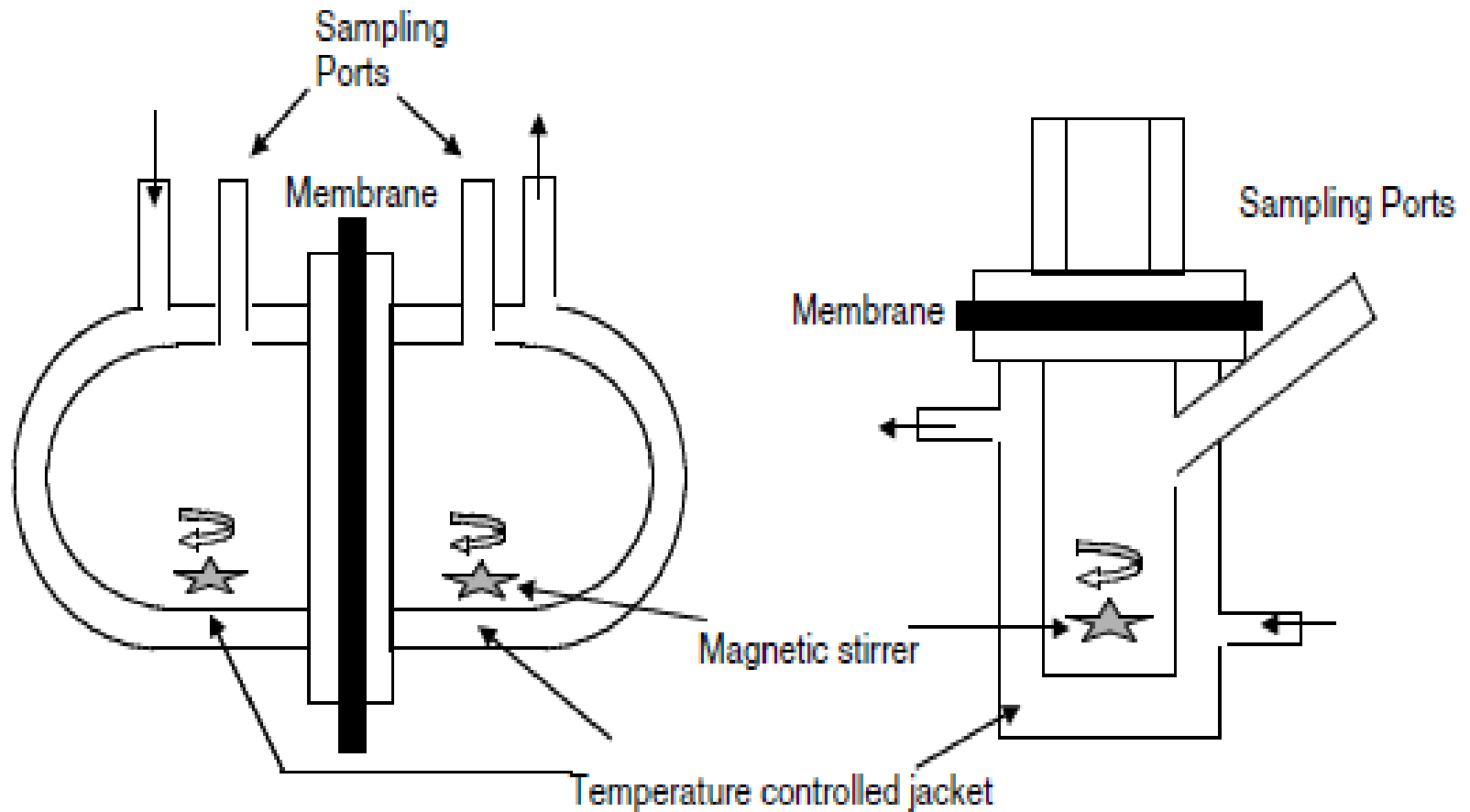


FIG. Side-by-side and vertical simple diffusion cells.

Diffusion study...

Plexiglas three compartment diffusion cell

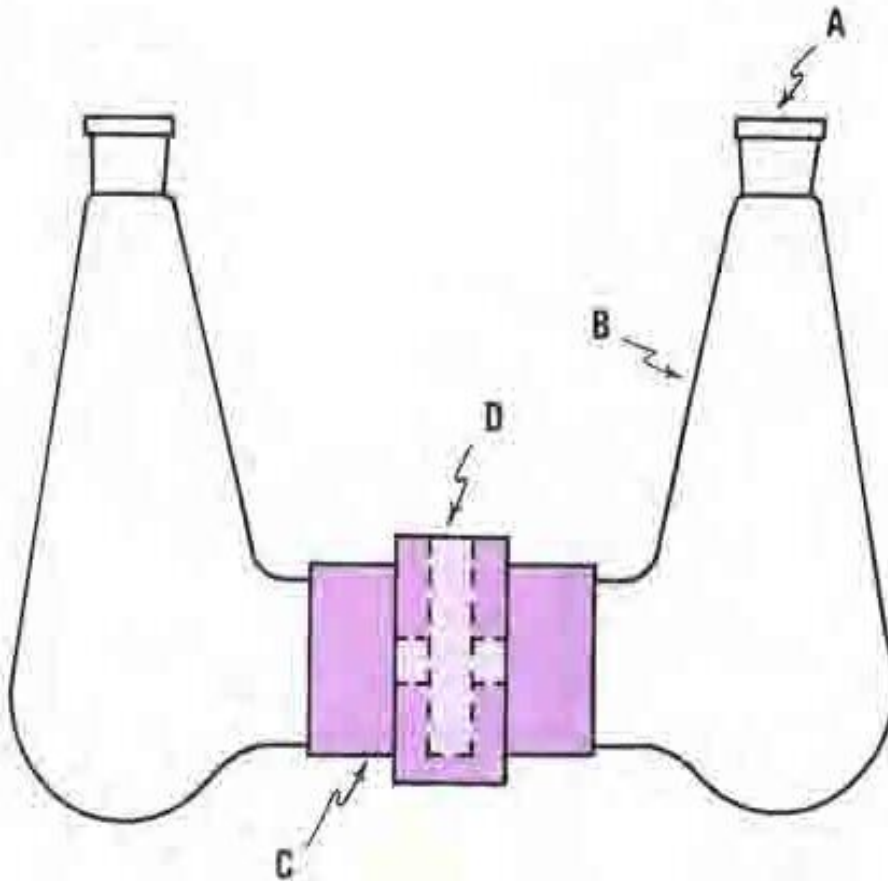
- This is three compartment diffusion cell for use with either synthetic or isolated biological membrane
- The drug is allowed to diffuse from the two outer donor compartments in to a central receptor chamber
- Results are reproducible
 - ✓ b/c of created greater membrane surface exposure and improved analytical sensitivity

Diffusion study...

A permeability cell

- Diffusion cell for studying diffusion through stratum corneum of various permeants including gases, liquids and gels
- It is kept at constant temperature and gently shaken in the plane of the membrane
- Samples are withdrawn from the receptor chamber at definite times and analysed for the permeant

Diffusion study...



A= glass stopper

B= glass chamber

C= aluminum collar

D= membrane and
sample holder

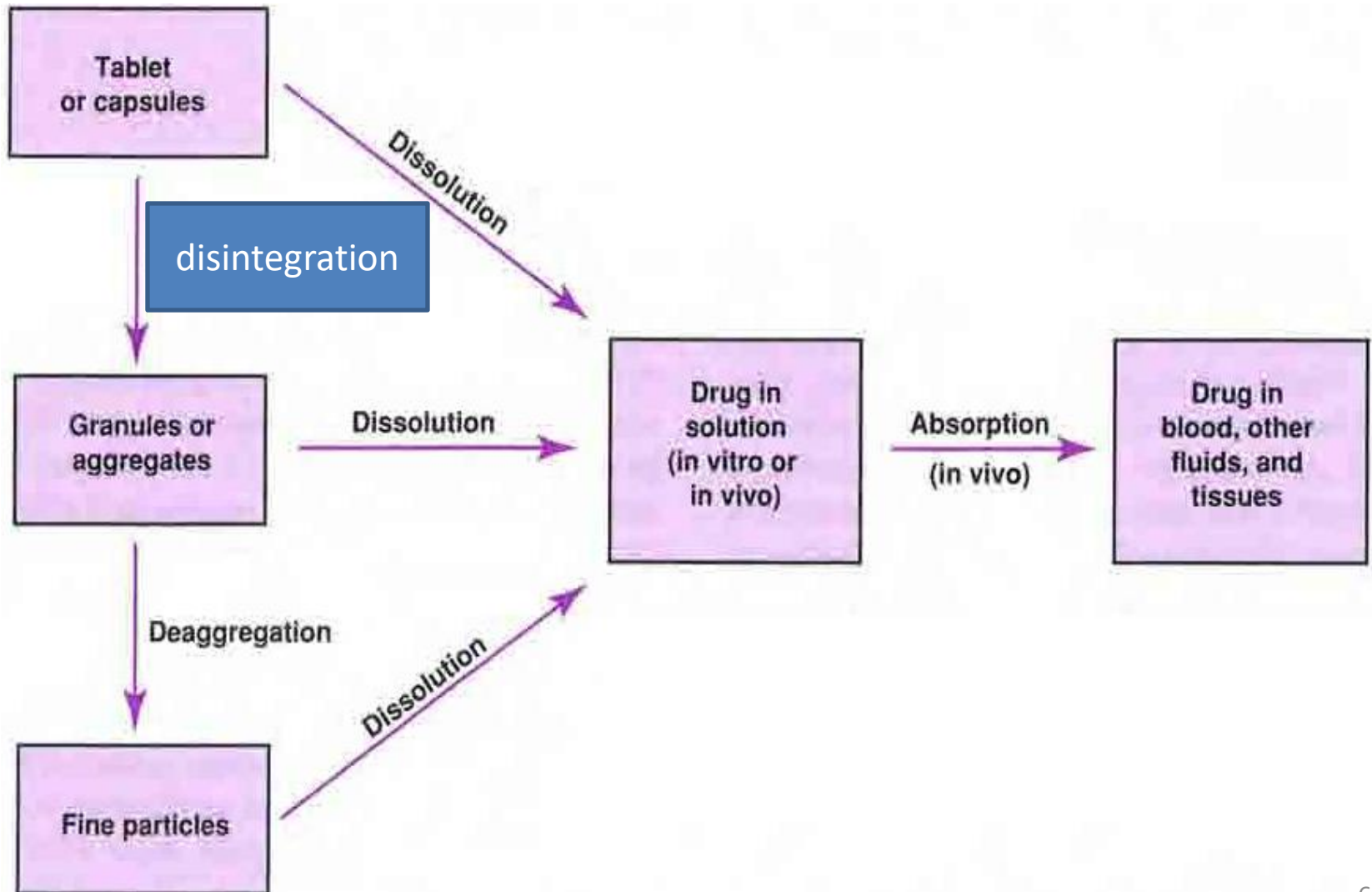
Dissolution of particles

- **Dissolution** is defined as the process by which solid phase (eg a tablet or powder) enters into solution phase. Or mass transfer from the solid surface to the liquid phase.
- **Rate of dissolution** is the amount of drug substance that goes in to solution per unit time under standardized conditions of temperature and solvent composition.

Dissolution of particles...

- When a solid dosage form such as a tablet is introduced into water, the drug contained in the tablet begins to pass into solution.
- Simultaneously, the solid matrix of the tablet disintegrate in to granules and these granules in turn deaggregate into fine particles
- Dissolution of the drug takes place from tablet, the disintegrated granules and from the fine particles.

Dissolution of particles...



Dissolution of particles...

- Although the effectiveness of the tablet in releasing the drug is somewhat dependent on the rate of disintegration and deaggregation, of more important is the rate of dissolution of the solid drug
- Most often the dissolution step is the slowest and hence the limiting or rate-controlling step in the bioabsorption of drugs of low solubility.

Dissolution of particles...

- According to dissolution theory, the dissolution of a solid in a liquid may be regarded as being composed of two consecutive stages
- 1. First is an interfacial reaction that results in the liberation of solute molecules from the solid phase.
 - involves a phase change, so that molecules of solid become molecules of solute in the solvent
 - The solution in contact with the solid will be saturated (because it is in direct contact with undissolved solid)
 - Its concentration will be C_s , a saturated solution

Dissolution of particles...

2. the solute molecules must migrate through the boundary layers surrounding the crystal to the bulk of the solution, at which time its concentration will be C .

- This step involves the transport of these molecules away from the solid-liquid interface into the bulk of the liquid phase under the influence of diffusion.
- Boundary layers are static or slow moving layers of liquid that surround all wetted solid surfaces.

Dissolution of particles...

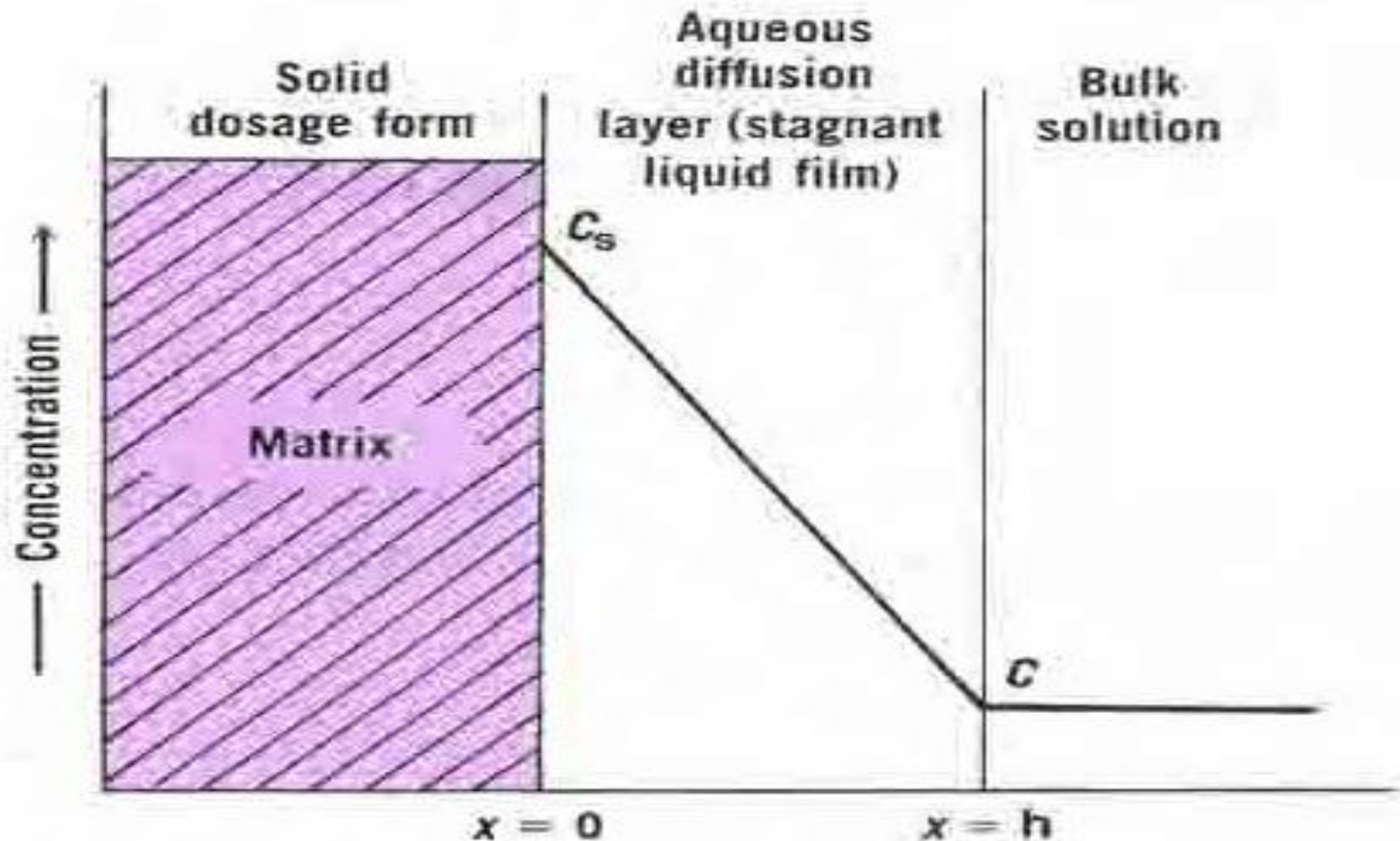
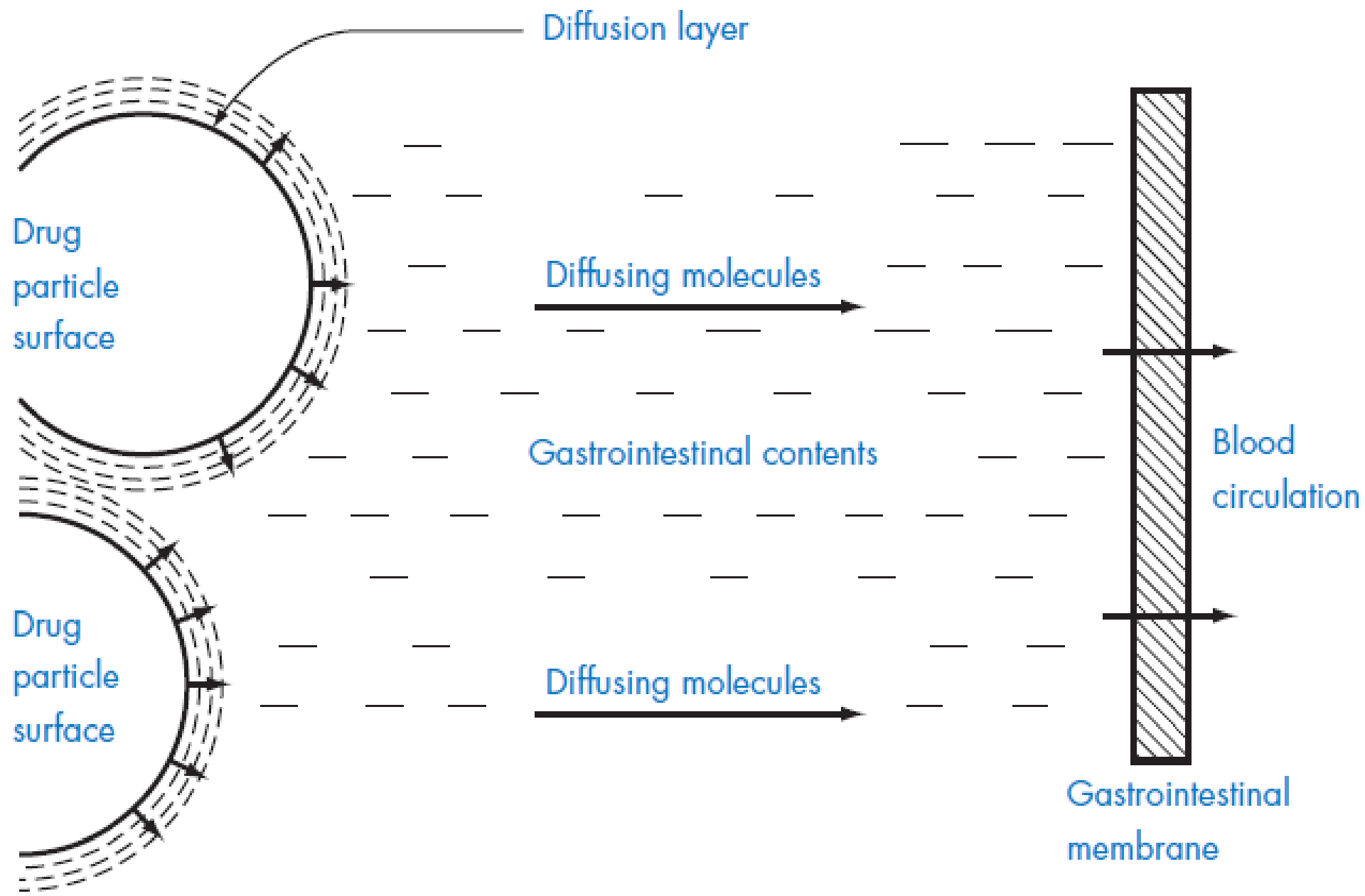


Fig: Dissolution of a drug from a solid matrix, showing the stagnant diffusion layer between the dosage form surface and the bulk solution

Schematic diagram of dissolution from a solid surface.



Dissolution of particles...

- Mass transfer takes place more slowly through these static or slow-moving layers, which inhibit the movement of solute molecules from the surface of the solid to the bulk of the solution.
- The concentration of the solution in the boundary layers changes from being saturated (C_s) at the crystal surface to being equal to that of the bulk of the solution (C) at its outermost limit.

Dissolution of particles...

- the rate of dissolution is determined by the rate of the slower step
 - Therefore diffusion of dissolved solute across the static boundary layer of liquid that exists at a solid-liquid interface is the rate limiting steps

Dissolution of particles...

- The rate at which a solid dissolves in a solvent is quantitatively given by the Noyes-Whitney equation

$$\frac{dM}{dt} = \frac{DS}{h}(C_s - C) \quad (11)$$

or

$$\frac{dC}{dt} = \frac{DS}{Vh}(C_s - C) \quad (12)$$

Where

M = the mass of solute dissolved in time t (usually mg or mmol)

dM/dt = the mass rate of dissolution (mass/time)

t = time (s)

Dissolution of particles...

D = diffusion coefficient of the solute (cm^2/s)

S = the surface area of the exposed solid (cm^2)

h = thickness of the diffusion layer (cm)

C_s = the solubility of the solid (i.e the concentration of a saturated solution of the compound at the surface of the solid and at the temperature of experiment)

C = concentrations of the solute at the bulk solution at time t

Dc/dt = the dissolution rate

V = the volume of the solution

Dissolution of particles...

- If the solute is removed from the dissolution medium by some process at a faster rate than it passes into solution or
- if the volume of the dissolution medium is so large that C is not allowed to exceed 10% of the value of C_s
- dissolution is said to occur under 'sink' conditions, and Eq. 11 become

$$dM/dt = DSC_s/h \quad (13)$$

Dissolution of particles...

- 'sink' conditions may arise in vivo when a drug is absorbed from its solution in the gastrointestinal fluids at a faster rate than it dissolves in those fluids from a solid dosage form (a tablet)
- If solute is allowed to accumulate in the dissolution medium to such an extent that the above approximation is no longer valid, i.e. when $C > C_s/10$, then 'non-sink' conditions are said to be in operation.
- When $C = C_s$, the overall dissolution rate will be zero
 - This occurs if dissolution medium is saturated with solute

Dissolution of particles...

Drawback of Noyes-Whitney equation

- In the derivation of Noyes-Whitney equation, it was assumed that h and S are constant
 - S decrease as particle size decrease when solid undergoes dissolution
 - Thickness of the static layer also altered by agitation

Dissolution of particles...

Problem 1: Calculate the mass rate of dissolution of a hydrophobic drug having the following physicochemical characteristics:

surface area = $2.5 \times 10^3 \text{ cm}^2$

saturated solubility = 0.35 mg/mL (at room temperature)

diffusion coefficient = $1.75 \times 10^{-7} \text{ cm}^2/\text{s}$

thickness of diffusion layer = $1.25 \text{ }\mu\text{m}$

conc of drug in bulk = $2.1 \times 10^{-4} \text{ mg/mL}$

Problem 2: What would the mass rate of dissolution be in Problem 1 if the surface area was increased to $4.3 \times 10^4 \text{ cm}^2$?

In-vitro dissolution testing methods

- Alternative to *in vivo* bioavailability determination
- The most widely used apparatus for dissolution study are rotating **Basket** and **paddle** type dissolution apparatus

Rotating Basket Apparatus (Apparatus 1)

- Consists of a 40 mesh stainless steel basket attached to the end of a stirring shaft which in turn is attached to a variable speed motor
- A tablet or a capsule is placed within the basket and immersed in the dissolution fluid contained in a cylindrical glass vessel with hemispherical bottom of one liter capacity which partially immersed in a water bath

In-vitro dissolution testing methods...

- The basket is located centrally in the vessel at a distance of 2 cm from the bottom and rotated by a variable speed motor through a shaft.
- The dissolution fluid is water, buffer solution or dilute acid simulating GI pH and maintained at 37 °c
- Samples are withdrawn at different times for analysis
- The volume the dissolution medium is maintained constant by replacing the amount removed with fresh dissolution fluid

In-vitro dissolution testing methods...

- All metal parts like basket and shaft are made of stainless steel.



In-vitro dissolution testing methods...

Rotating Paddle Apparatus (Apparatus 2)

- Here, basket is replaced with a paddle.
- The position and alignment of the paddle are specified in the official books.
- During the operation, the tablet is dropped in the dissolution fluid



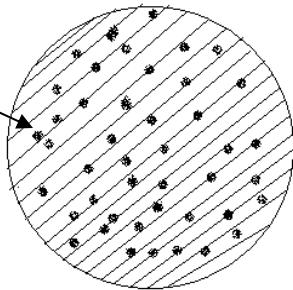


Significance of diffusion

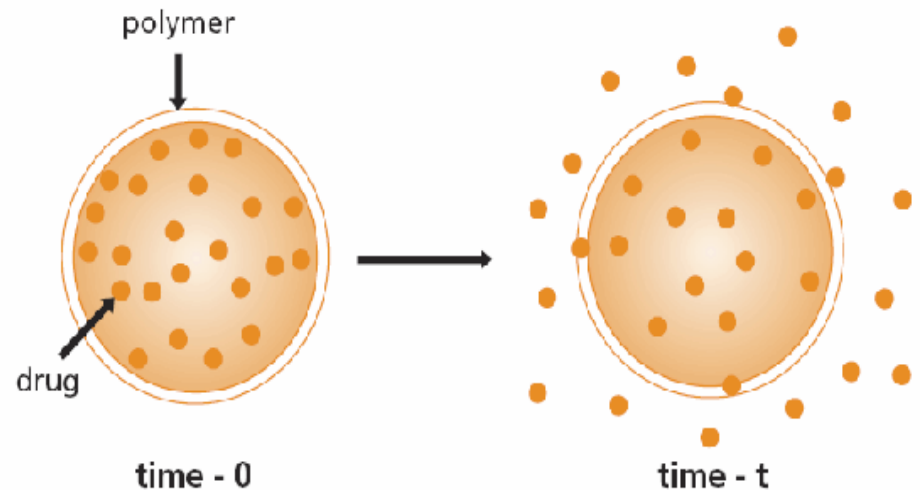
Diffusion Controlled drug Delivery system

- Polymers function as physical barriers to drug transport and hence control the rate of release of a drug when administered orally.
- Diffusion controlled systems are of two types:
 - a. reservoir devices
 - b. matrix devices.

Soluble drug



matrix devices.



reservoir devices

Significance of diffusion...

Reservoir devices

Drug release involves the following processes:

- ✓ GI fluid penetrate the coating and the drug dissolves to produce a saturated solution
- ✓ the drug diffuses through the polymeric coating
- A fairly constant rate of drug release occurs if the concentration of drug in the core remains at saturation
- The release of solutes from polymeric drug delivery systems are based on **Fick's first law of diffusion.**

Significance of diffusion...

Matrix Devices

- A matrix (monolithic) device consists of an inert polymeric matrix in which a drug is uniformly distributed.
 - ✓ Drugs can be dissolved in the matrix
 - ✓ the drugs can be present as a dispersion
- The state of presentation leads to different release characteristics.

Significance of diffusion...

Dissolved Drug

- When the system is a homogeneous matrix containing a dissolved drug, the release can be described by a solution to **Fick's second law of diffusion**

Dispersed Drug

- If a drug is dispersed as a solid in the homogeneous matrix (monolith) instead of being completely dissolved, the release kinetics are altered
- The derivation of the release rate expression is based on **Fick's first law of diffusion**

Significance of diffusion...

- Higuchi developed a model for the analysis of a homogeneous polymer matrix containing a suspended drug.
- The expression of the amount of drug released per unit area of the matrix at time t

$$Q = [D(2A - C_s)C_s t]^{1/2} \quad (14)$$

- The instantaneous rate of release of a drug at time t is obtained by differentiating eq. 14

$$\frac{dQ}{dt} = \frac{1}{2} \left[\frac{D(2A - C_s)C_s}{t} \right]^{1/2} \quad (15)$$

Significance of diffusion...

- where Q is the amount of drug released per unit area in time t
 D and C_s are the diffusion coefficient and solubility of the drug in the polymeric matrix, respectively

A is the concentration of drug initially present in the matrix

dQ/dt is instantaneous rate of release of a drug at time t

- When $A \gg C_s$,
eq 14 reduces to

$$Q = (2ADC_s t)^{1/2} \quad (16)$$

eq 15 reduces

$$\frac{dQ}{dt} = \left(\frac{ADC_s}{2t} \right)^{1/2} \quad (17)$$

Significance of diffusion...

- Q and dQ/dt , can be altered by increasing or decreasing the drug solubility, C_s , in the polymer matrix by complexation
- The total concentration of A of the drug also affect the rate of drug release

Significance of diffusion...

Problems

1. What is the amount per unit area, Q , released from a tablet matrix at time $t=120$ min and instantaneous rate of drug release occurring at this time ? the total concentration of drug in the homogenous matrix, A , is 0.02 g/cm^3 . The drug solubility is $1.0 \times 10^{-3} \text{ g/cm}^3$ in the polymer matrix. The diffusion coefficient, D , of the drug in the polymer matrix at 25°C is $6.0 \times 10^{-6} \text{ cm}^2/\text{sec}$.

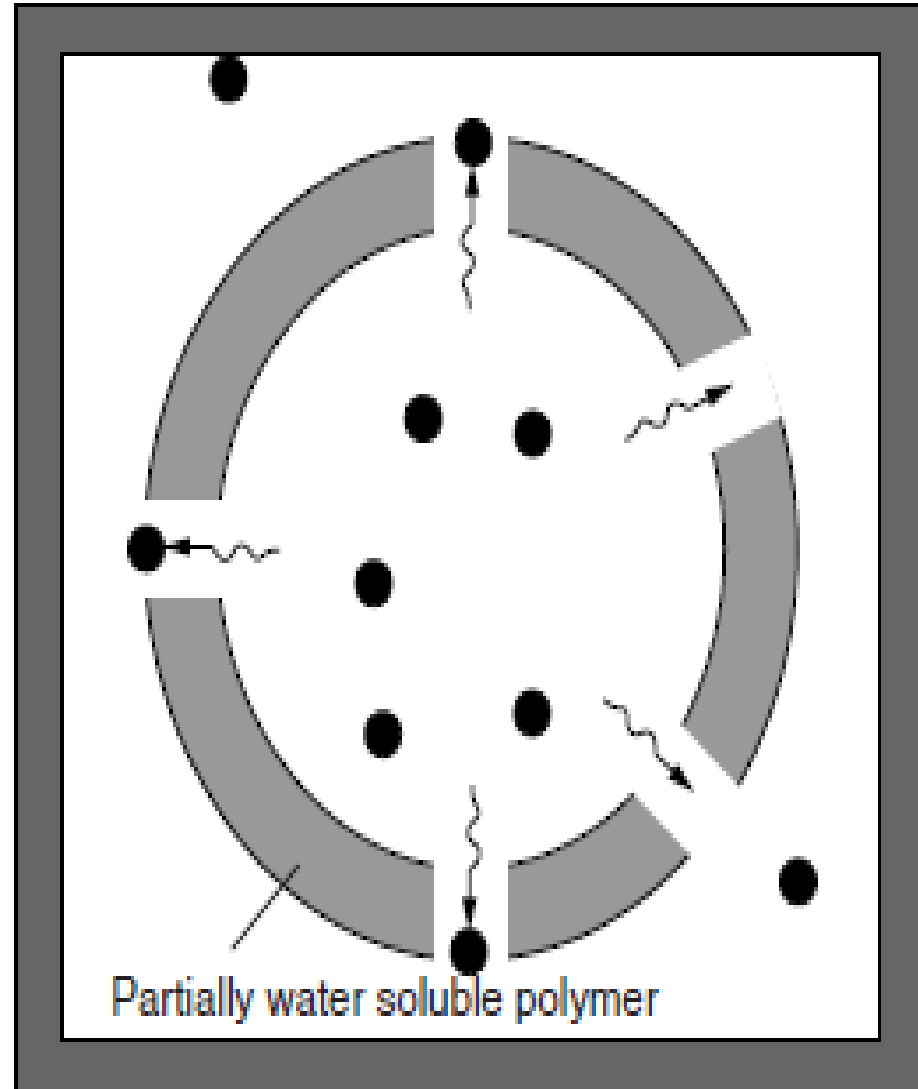
Significance of diffusion...

Dissolution and Diffusion Controlled Systems

- In this system, the drug forms **the core**, which is surrounded by a partially **water-soluble polymer**.
- Drug release is triggered by the dissolution of the water-soluble part of the polymer membrane
 - diffusion of the drug through the holes or pores in the polymer.
- achieved by incorporating water-soluble additives in the polymer coating.

Significance of diffusion...

- When placed in an aqueous environment, the water-soluble additives (pore-formers) in the polymer coating dissolve to leave pores.



Significance of diffusion...

- The kinetics of the release of drugs from dissolution-controlled systems is often described by the Noyes-Whitney equation:

$$dC/dt = DS/hV (C_s - C) \quad (18)$$

Where:

dC/dt is the dissolution rate;

V is the volume of the solution;

D is the diffusion coefficient,

h is the thickness of aqueous diffusion layer

S is the surface area of the exposed solid.

C_s is the saturated solubility of the drug,

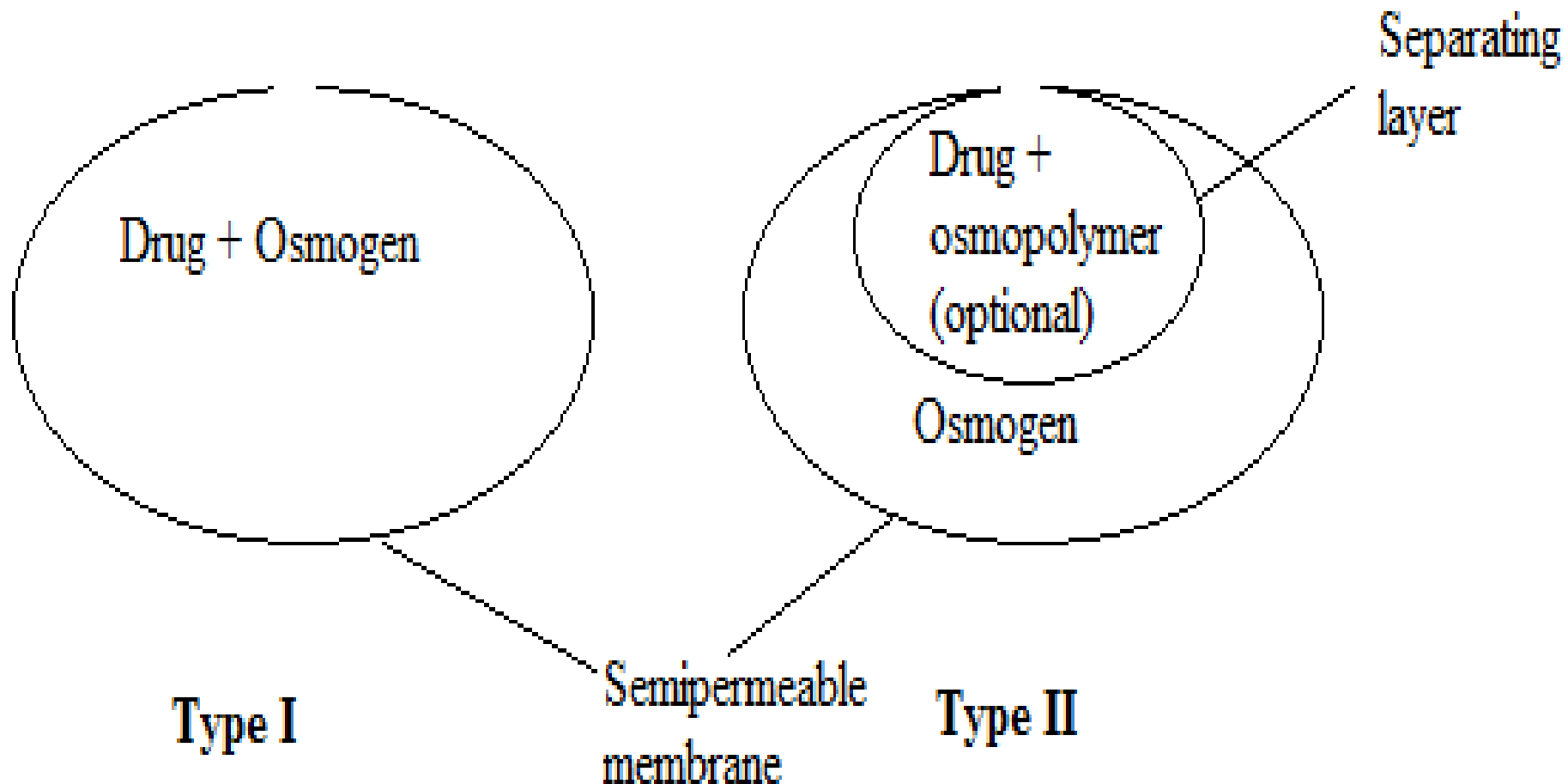
C is the concentration of the drug in the bulk solution.

Significance of osmosis

Osmotic controlled drug release

- This type of delivery device has a **semipermeable** membrane that allows a controlled amount of water to diffuse into the core of the device filled with drug and osmotic agent
- A water-sensitive component in the core can either dissolve or expand to create **osmotic pressure**
 - ✓ push the drug out through a small **delivery orifice**

Significance of osmosis...



Significance of osmosis...

- The main advantage of the osmotic pump system
 - constant release rate can be achieved, since it relies simply on the passage of water into the system
- The release rate of the device is affected by:
 - the amount of osmotic agent
 - surface area and thickness of semipermeable membrane
 - the size of the hole.

Significance of osmosis...

- The rate at which drug is pumped out of the osmotic system through the orifice is equal the volume flow rate of water into the core multiplied by drug concentration at saturation

$$dM/dt = (dV/dt) C_s \quad (19)$$

dM/dt is the amount of drug (mg) delivered per unit of time (hours)

dV/dt is the volume flow rate [milliliters per unit of time (hours)]

C_s is the concentration of the saturated solution

Significance of osmosis...

- The volume flow rate of water into the osmotic device can be expressed in terms of:

membrane permeability (P),
semipermeable membrane surface area (A),
thickness (h) of the membrane,
osmotic pressure difference ($\Delta\pi$),

$$dV/dt = (AP/h)(\Delta\pi) \quad (20)$$

Therefore, $dM/dt = (AP/h)(\Delta\pi) C_s \quad (21)$